FROM TREES TO TECH · ACTIVITY TTT-10 · DATA + CLIMATE

Tree Ring Climate Detective

Read the rings to reconstruct the climate events a tree lived through.

Win condition: Most accurate climate-event inferences with claim-evidence-reasoning.

THE SCIENCE YOU ARE USING

Rings as proxy data. A tree adds one ring per year. Wide rings mean a good growing season; narrow rings mean stress like drought, cold, or crowding. Because the tree cannot record numbers, the rings are a proxy: an indirect record of past climate.

Claim, evidence, reasoning. You infer events from ring patterns, then justify each call with the specific rings that support it. Scars, sudden narrowing, and runs of wide rings are your clues. Strong inferences cite the evidence.

HOW TO BUILD AND RUN IT

01 **Learn the ring code.** Wide equals good growth, narrow equals stress, scar equals fire or injury.

02 **Read the sequence.** Work along the rings from center to bark, noting each pattern and the year it maps to.

03 **Infer the events.** Match patterns to likely events: drought, fire, a crowded stand, a recovery.

04 **Cite your evidence.** For each inference, point to the exact rings that justify it on a claim-evidence card.

05 **Synthesize the story.** Combine your inferences into one short climate history for the tree.

Safety: Sand wood cookies smooth if used. No cutting tools with students. Allergy: Wood dust sensitivity note if using real samples.

MATERIALS

Printed ring cards	set
Wood cookies or images	shared
Answer boards	per team
Timers	per team
Claim-evidence cards	per team

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ____________________

____

The ONE variable we are testing: ____________________

Baseline result: ____________________

After redesign: ____________________

DEFEND YOUR DESIGN

What evidence made you change your design? ____________________

____

If you ran it again, what is the ONE thing you would change? ____________________

____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Correct inferences	40
Evidence justification	20
Speed	20
Final synthesis	20
Total	**100**